

Abstract

- A web-based platform for conducting online examinations, managing students and exams, evaluating submissions, and supporting browser-based AI-assisted proctoring.
- Outcome: centralized examination management with automated backend scoring and browser-based monitoring indicators.

Problem Statement & Objectives

- Problem: Online examinations need reliable exam delivery, automated evaluation, secure data management and useful monitoring.
- Objective 1: Provide student and admin portals for examination workflows.
- Objective 2: Store exams, questions, submissions and requests in MongoDB.
- Objective 3: Assist proctoring through browser camera-based object detection.
- Scope: Authentication, exams, questions, submissions, results and proctoring indicators.

Existing vs Proposed System

Feature	Existing System	Proposed System
Exam Setup	Manual/basic	Centralized admin
Questions	One-by-one entry	Manual + file import
Evaluation	Manual/basic	Backend scoring
Monitoring	Limited	AI-assisted indicators
Results	Basic records	Stored + admin review

Feasibility Study

- Technical: Uses established React, FastAPI, MongoDB, JWT and browser-AI technologies.
- Economic: Uses open-source software and existing development resources.
- Operational: Provides separate student and admin workflows with a browser-based exam interface.
- Schedule: Modular components support development and testing within the semester.

Results & Testing

- JWT login and protected API access tested.
- MongoDB connectivity, exam/question/submission flow and backend scoring tested.
- Browser proctoring tested for person and mobile-phone detection.

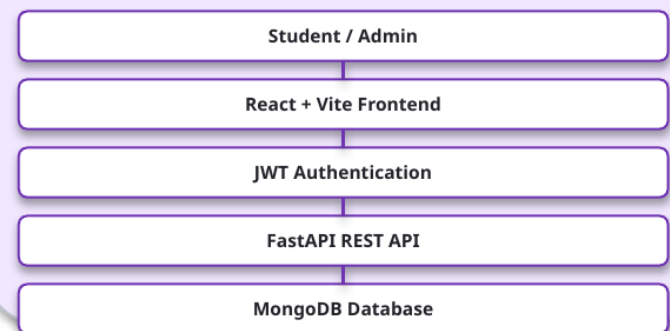
Conclusion

- Provides a centralized online examination workflow with authentication, exam management, backend evaluation, MongoDB persistence and AI-assisted browser proctoring.
- Separates frontend, API, database and monitoring responsibilities.

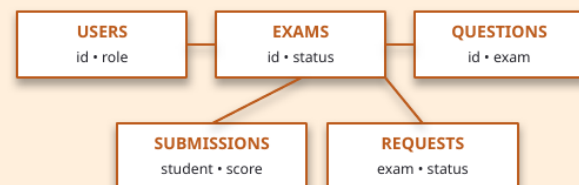
Tools & Technologies

- Front-End: React 19, Vite, React Router, Tailwind CSS
- Back-End: Python FastAPI + REST APIs
- Database: MongoDB
- Authentication: JWT + bcrypt
- Modules: Online Examination, Exam Management, Automated Evaluation, AI-Assisted Proctoring
- Tools: Git & GitHub, VS Code, Postman

System Architecture

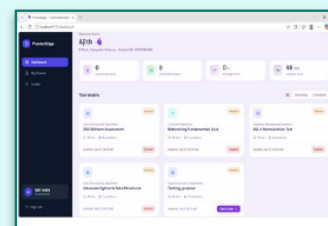


ER Diagram & Database Design

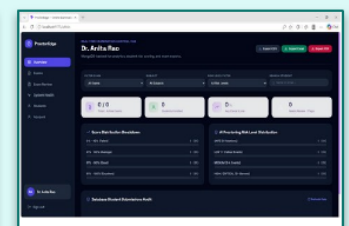


MongoDB collections: users, exams, questions, submissions, requests.

Implementation Screenshots



Student Dashboard



Admin Dashboard – Dr. Anita Rao

